

Hypotheses & Examples

Business Analyst Case Study | Dataset: Olist E-Commerce (seller marketplace)

02 · HYPOTHESES & EXAMPLES

Approach

Rather than exploring the data without direction, three specific hypotheses were formed up front, each representing a plausible operational explanation for order cancellations. Each was tested independently against the data before being accepted, rejected, or qualified.

#	Hypothesis	Rationale
H1	Product category drives cancellation rate	Some product types may be inherently harder to fulfill (fragile, high-return-risk, or availability-sensitive categories).
H2	Seller tenure drives cancellation rate	Newer sellers may be less operationally mature — less accurate stock counts, slower fulfillment, less experience with the platform's requirements.
H3	Delivery delay drives cancellation rate	Customers may cancel orders themselves when deliveries run late, or system logic may auto-cancel severely delayed orders.

H1 — Product Category

Test: Group orders by product category, compute cancellation rate per category, filtered to categories with 50+ orders to avoid small-sample distortion.

Result: Weak, mostly volume-driven signal. The highest-rate category among small-volume groups (DVDs/Blu-ray) showed a 3.39% rate, but on only 59 orders. Among higher-volume categories, rates ranged narrowly (roughly 0.2%–0.9%), without one category standing out as a structural driver.

Example — why small samples mislead: DVDs/Blu-ray showed 2 cancellations out of 59 orders = 3.39%. If just one fewer order had been cancelled (1 of 59), the rate would drop to 1.7% — nearly half. A rate this sensitive to a single data point is not a reliable signal, which is exactly why the 50-order minimum filter was applied before ranking categories.

H2 — Seller Tenure

Test: Bucket sellers by days since their first recorded order (0–30 days, 30–90 days, 90+ days), then compute cancellation rate per bucket.

Result: Confirmed, and the strongest finding in this analysis. New sellers (0–30 days) showed a 1.08% cancellation rate versus 0.40% for established sellers (90+ days) — more than double, with a clear step-down through the mid-tenure bucket (0.52%), consistent with a genuine operational maturation curve rather than a random split.

Example — ruling out a category confound: within the automotive category specifically, new sellers showed a 2.51% cancellation rate versus 0.51% for established sellers in that same category. Because the gap holds within a single category, it cannot be explained away as "new sellers happen to sell riskier products" — tenure itself, not category

mix, is driving the difference.

H3 — Delivery Delay

Test: Bucket delivered orders into early / on-time / late based on estimated vs. actual delivery date, then compare cancellation rate across buckets.

Result: Ruled out. Cancelled and unavailable orders almost never reach a delivery date at all — they fail before that stage of the order lifecycle. As a result, this comparison structurally excludes almost all of the "problem" orders from both buckets, making the two groups incomparable for this purpose.

Example: the observed rates for both buckets came out at approximately 0.01%, essentially flat. This is not evidence that delivery delay has no effect on cancellations in reality — it is evidence that this data cut cannot measure that effect, since the orders that would show the effect (the cancelled ones) are filtered out before the comparison even begins. This distinction — "not measurable here" vs. "disproven" — is treated as an important intellectual honesty point rather than glossed over.

Summary of Hypothesis Outcomes

Hypothesis	Outcome	Confidence
H1: Category	Weak / volume-sensitive signal only	Low
H2: Seller tenure	Confirmed — strongest driver found	High (statistically validated)
H3: Delivery delay	Ruled out — not measurable with this data cut	N/A